"Is This Safe?" — Volunteer Answer Key

FOOD SAFETY

FS-K01 **SAFE** *Kids (under 10)*

You washed your hands with soap and warm water before helping make a sandwich. You also wiped the kitchen counter with a clean cloth before placing the bread down.

Why: Washing your hands removes germs that can make you sick. Cleaning the counter helps stop germs from spreading to your food. This way, your sandwich stays safe to eat.

Volunteer notes: Good personal hygiene and kitchen cleanliness are essential for food safety. Washing hands for 20 seconds with soap and cleaning food surfaces prevents contamination from bacteria such as Salmonella or E. coli .

Source: SP 50-1000 Turkey basics - "Wash your hands before preparing food. Clean kitchen surfaces with soap and water and sanitize to prevent harmful bacteria."

FS-K02 **NOT SAFE** *Kids (under 10)*

You found some jam in the fridge with fuzzy mold growing on the top. You cut off the fuzzy-looking part and thought it was okay to eat the rest.

Why: Mold can grow roots deep inside the jam, not just on the fuzzy part you see. These molds might make you sick, so it's best to throw the jam away if you see mold.

Volunteer notes: Molds can produce poisonous substances called mycotoxins that spread throughout the food, making it unsafe even after cutting off visible mold. All moldy jams should be discarded to prevent illness .

Source: SP 50-1000 Turkey basics - "Molds have roots that invade food deeply, and toxins may spread throughout, making moldy food unsafe to eat."

FS-K03 **IT DEPENDS** *Kids (under 10)*

You left some potato salad out on the kitchen table for a while during a family party. You wonder if it's okay to eat it now.

Why: Potato salad should not be left out too long because warm temperatures make bacteria grow fast. If it was out for more than 2 hours or your home is very warm, it's safer to throw it away. If it was less time and feels cool, it might still be safe.

Volunteer notes: Perishable foods, especially protein-rich ones like potato salad, must not be left out longer than 2 hours at room temperature (less in hot weather). This prevents bacteria growth that can cause illness .

Source: SP 50-1000 Turkey basics - "Discard any perishable food left at room temperature more than 2 hours. Protein-rich foods need careful handling to prevent rapid bacteria growth."

FS-W01 **SAFE** *Tweens (10-14)*

You just washed your hands with soap and water for 20 seconds before making a peanut butter sandwich. You also wiped down the countertop with a clean cloth. Is this safe to eat?

Why: Washing your hands well and cleaning surfaces with soap helps remove harmful germs. This stops bacteria or viruses from getting into your food. This is one of the best ways to keep food safe and prevent food poisoning.

Volunteer notes: Proper handwashing for 20 seconds with soap and cleaning kitchen surfaces prevents contamination from harmful bacteria like Salmonella and E. coli. This is a key food safety practice recommended by USDA and OSU Extension to avoid foodborne illness.

Source: SP 50-1000 Turkey basics - "Wash your hands with soap and warm water and rub thoroughly for 20 seconds... Clean and sanitize food preparation surfaces often with hot, soapy water... Clean hands can save lives."

FS-W02 **NOT SAFE** *Tweens (10-14)*

A hot dog was left out on the picnic table at a summer party for 3 hours in 85°F weather before people ate it. Is it safe to eat?

Why: Leaving meat like hot dogs in warm weather for more than 2 hours lets bacteria grow fast. This food is in the "danger zone" temperature where germs can make you sick. Eating it can cause food poisoning.

Volunteer notes: Perishable foods like cooked meat left out longer than 2 hours at temperatures between 40°F and 140°F can grow dangerous bacteria rapidly. This violates the 2-hour rule from USDA and OSU Extension summer food safety guidelines.

Source: SP 50-1000 Turkey basics - "Never leave perishable food out for more than two hours (or one hour if it is hotter than 90°F outside)...

warm temperatures promote rapid growth of microorganisms."

FS-W03 **IT DEPENDS** *Tweens (10-14)*

Your family found a jar of homemade pickles in the pantry that's been opened and left out at room temperature for a few days. Is it safe to eat?

Why: Pickles are acidic foods, which usually stop dangerous bacteria from growing. However, if the pickles were not made with enough vinegar or the jar was contaminated, they might not be safe. Keep pickles refrigerated after opening to be safe.

Volunteer notes: Pickles are acidic (pH below 4.6), which typically prevents harmful bacteria growth. Safety depends on proper acid level and storage. Unrefrigerated opened pickles may pose a risk. Follow safe canning and storage practices per USDA and OSU Extension guidance.

Source: SP 50-1000 Turkey basics - "Acidic foods include all fruit, tomatoes, and pickles made with an adequate amount of vinegar. Low acid foods are more of a concern. Temperature and time outside refrigeration affect safety."

FS-T01 **SAFE** *Teens (14-18)*

You packed a picnic with cooked chicken, potato salad, and fresh fruit. You kept the cooler closed and in the shade with plenty of ice packs. You served and ate all the food within two hours.

Why: Keeping perishable foods like cooked chicken and potato salad cold (below 40°F) and out of the danger zone (40-140°F) slows harmful bacteria growth. Serving within two hours ensures safety. The cooler being in the shade with ice packs maintained the needed cold temperature.

Volunteer notes: The 2-hour rule and keeping perishable foods cold in an insulated cooler prevents rapid bacterial growth that causes foodborne illness. The critical point is time and temperature control to keep food safe during picnics or outdoor events.

Source: SP 50-1000 Turkey basics - "Adhere to the 2 hour rule: discard any perishables left at room temperature for more than 2 hours. Keep perishable food cold until you're ready to serve it. When possible, put the cooler in the shade and keep the lid on."

FS-T02 **NOT SAFE** *Teens (14-18)*

A hamburger cooked on a barbecue was left out on the kitchen counter for 5 hours before being eaten.

Why: Cooked meat left at room temperature for more than two hours is unsafe because harmful bacteria grow quickly in the 'danger zone' between 40°F and 140°F. Eating this hamburger could cause food poisoning, as bacteria like Salmonella multiply rapidly.

Volunteer notes: Food safety standards state perishable foods should not be at room temperature over 2 hours (1 hour if over 90°F). This prevents bacterial growth causing illness. A likely follow-up: 'How can I tell if food is spoiled?' - Often it looks and smells normal, so time and temperature rules are key.

Source: SP 50-1000 Turkey basics - "Discard any perishables left at room temperature for more than 2 hours. Moist, protein-rich foods need careful handling as bacteria multiply rapidly in the danger zone."

FS-T03 **IT DEPENDS** *Teens (14-18)*

You want to freeze fresh vegetables for future use. You blanched them briefly in boiling water, then froze them immediately. However, the power went out overnight and the vegetables thawed partially before the power returned.

Why: Blanching then freezing vegetables slows bacterial growth, making freezing safe. But if vegetables thaw and warm above 40°F for too long, bacteria can grow and make them unsafe. If they still have ice crystals when power returns and were kept cold enough less than two days, refreezing can be safe.

Volunteer notes: Blanching kills enzymes; freezing preserves quality and safety. The safety risk depends on temperature and time during the power outage. When above 40°F too long, bacterial growth risks make refreezing unsafe. Ask: 'How long was the power out? Did the vegetables get warmer than 40°F?'

Source: PNW 214 Freezing fruits and vegetables - "If the temperature has warmed above 40°F, foods may not be safe for refreezing. Check your freezer thermometer... Do not eat thawed vegetables that are above 40°F."

FS-A01 **SAFE** *Adults*

You pack a cooler with freshly cooked chicken, keep it chilled below 40°F with ice packs during a summer picnic, and consume all leftovers within 2 hours of serving.

Why: Keeping perishable cooked foods like chicken chilled below 40°F and consuming them within 2 hours prevents harmful bacterial growth. This practice fits the 2-hour rule recommended for food safety in warm conditions, especially important for protein-rich foods. Proper cooling and timely consumption are key to preventing foodborne illness.

Volunteer notes: OSU Extension and USDA recommend keeping perishable foods below 40°F and following the 2-hour rule for safe consumption at events like picnics, especially in warm weather. Follow-up Q: What if food is left out longer than 2 hours? That food should be discarded to prevent risk of illness.

Source: SP 50-814 Summer food safety - "Adhere to the 2 hour rule: discard any perishables left at room temperature for more than 2 hours. Moist, protein-rich foods need careful handling."

FS-A02 NOT SAFE *Adults*

You find a can of home-canned green beans with a bulging lid and a foul odor inside the jar after storage at room temperature for several months.

Why: A bulging lid and off odors in home-canned low-acid foods signal possible bacterial growth including dangerous botulism toxin. Such cans must never be opened or consumed. These are clear signs of spoilage and pose a serious health risk.

Volunteer notes: USDA and OSU Extension warn that bulging lids indicate gas production by bacteria, including Clostridium botulinum. Any home-canned food showing this must be discarded immediately. Follow-up Q: Can heating or tasting test safety? No, always discard suspicious canned goods without tasting.

Source: SP 50-490 Botulism - "Bulging cans and bad odors are signs of spoilage caused by dangerous bacteria such as botulism."

FS-A03 IT DEPENDS *Adults*

You plan to can tomato salsa at home without adding lemon juice or citric acid, using a boiling water bath canner at your local altitude of 4000 feet.

Why: Safety hinges on the acidity level and processing adjustments for altitude. Tomatoes are borderline in acidity; without added acid like lemon juice or citric acid, the salsa may be unsafe in a boiling water bath. Also, altitude requires longer processing time or higher pressure to destroy pathogens. Without proper acidification and altitude adjustment, canned salsa may be unsafe.

Volunteer notes: USDA guidelines require acidifying tomato products when boiling water canning and increasing processing time per altitude. Follow tested recipes specifying acid additions and adjusted process times for your elevation. Follow-up Q: What if I don't add acid? The risk of botulism and spoilage increases significantly.

Source: SP 50-1001 PH values - "Low acid foods (pH above 4.6) require pressure canning; adding acid to tomato products is necessary for safety in boiling water canning; altitude affects processing time."

CANNING FRUITS

CF-K01 SAFE *Kids (under 10)*

You helped Mom wash fresh peaches and packed them tightly in clean jars. After adding the right amount of syrup and sealing the jars with new lids, you boiled them in a water bath canner just like the recipe said.

Why: This is safe because the peaches were fresh, clean, and packed in jars with new lids. The jars were boiled in a water bath canner, which cooks the fruit enough to keep germs away and seal the jars properly.

Volunteer notes: Fruits like peaches are high-acid foods and can be safely canned in a boiling water canner when following tested recipes precisely, including using new lids and proper processing times.

Source: PNW 199 Canning fruits - "When canning fruits, you can safely use a boiling water canner. Molds, yeasts, and bacteria which can grow in these high-acid foods are destroyed at boiling water-bath temperatures."

CF-K02 NOT SAFE *Kids (under 10)*

You found a jar of strawberry jam that had a fuzzy, green mold growing on top. The jar lid was still sealed tight, but the jam looked funny.

Why: Even if the jar lid is sealed, mold means tiny germs grew inside and the jam is spoiled. Eating moldy jam can make you sick, so it's not safe. Always throw away jams with mold.

Volunteer notes: Visible mold growth on home-canned fruits indicates spoilage and potential toxin presence. Foods showing mold should never be eaten or tasted. The seal does not guarantee safety if mold is present.

Source: Troubleshooting canning acid foods - "Spoilage could result if jars don't seal, and food is wasted."

CF-K03 IT DEPENDS *Kids (under 10)*

You canned peaches but left a jar on the kitchen counter overnight before putting it in the boiling water bath canner. The lid didn't pop down after cooking. Is it safe to eat?

Why: Leaving fruit out before canning can let germs grow. If the jam smells bad or the lid is still popped up after cooking, it's not safe to eat. But if it smells fresh, has no mold, and the jar is properly processed and sealed, it might be okay. When in doubt, throw it out!

Volunteer notes: Food left out before canning may develop spoilage microorganisms. Proper boiling water canning kills most germs, but unsealed jars or those with signs of spoilage should be discarded. Checking for bad smell and mold are key sensory checks at home for borderline cases.

Source: PNW 199 Canning fruits - "Work quickly throughout preparation and canning. If food is allowed to stand, quality is lowered and food spoilage is more likely to occur."

CF-W01 SAFE *Tweens (10-14)*

Maya carefully selects fresh, firm peaches and follows the exact tested recipe to can them in a boiling water canner. She uses new lids, fills the jars with boiling hot fruit, and processes them for the correct time. After cooling, she checks that the lids have sealed tightly before storing them in her cool pantry.

Why: Maya followed a tested recipe and used a boiling water canner because fruits are acidic enough for that method. Using new lids, filling jars hot, and processing for the full time kills harmful bacteria. Testing the seals makes sure no air or bacteria got in. This makes the peaches safe to store and eat later.

Volunteer notes: According to OSU/USDA standards, fruits are high-acid foods that can be safely processed in a boiling water canner. Using sterilized jars, new lids, and exact times ensures safety and quality. Proper cooling and seal checks prevent spoilage and botulism. Follow PNW 199 Canning fruits for tested recipes and times.

Source: PNW 199 Canning fruits - "Following the directions exactly is vital to ensure a product that is top quality and safe to eat. When canning fruits, you can safely use a boiling water canner. ... Do not use lids that are old, dented, or used before. Spoilage could result if jars don't seal, and food is wasted."

CF-W02 NOT SAFE *Tweens (10-14)*

Jake puts fresh strawberries into jars but uses old, dented lids he found in the cupboard from last year. He processes the jars in boiling water for less time than the recipe says. After cooling, some jars do not seal properly, but Jake decides to store them anyway.

Why: Using old or dented lids can prevent jars from sealing properly, allowing bacteria or molds to grow. Processing for less time can leave harmful microorganisms alive. Unsealed jars may spoil quickly and could cause food poisoning if eaten.

Volunteer notes: Old or damaged canning lids fail to create airtight seals, increasing spoilage and risk of pathogens like molds or bacteria. Undercooking by shortening processing time means bacteria or yeasts may survive. Always use new lids and follow processing times precisely for safety. PNW 199 Canning fruits warns about old lids and adequate processing.

Source: PNW 199 Canning fruits - "Do not use lids that are old, dented, deformed, or have been used before. Spoilage could result if jars don't seal, and food is wasted. Follow the preparation instructions carefully. Begin counting the processing time when the water comes back to a rolling boil."

CF-W03 IT DEPENDS *Tweens (10-14)*

Leah canned Asian pears using a boiling water canner but forgot to add lemon juice to the jars before processing. She isn't sure if the pears will be safe to store at room temperature or if she should keep them in the refrigerator.

Why: Asian pears are low-acid fruits and need added acid (like lemon juice) during canning to make the environment safe for boiling water canning. If lemon juice was forgotten, bacteria like Clostridium botulinum could survive and grow, making the pears unsafe if stored unrefrigerated. If stored cold (in the fridge), the risk is much lower, but it's best to follow tested recipes and add acid for room-temperature storage safety.

Volunteer notes: Some fruits, like Asian pears, require the addition of acid for safe boiling water canning. Without acid, dangerous bacteria can survive in jars sealed at boiling water canning temperatures. Storing without acid and without refrigeration can be unsafe. Follow PNW 199 guidelines to add acid for low-acid fruits processed in boiling water canners.

Source: PNW 199 Canning fruits - "Some fruits, such as Asian pears and figs, require the addition of acid (i.e., lemon juice) to be safely processed in a boiling water canner. These conditions also retard other undesirable changes such as vitamin loss, darkened color, and off-flavors."

CF-T01 **SAFE** *Teens (14-18)*

You prepare fresh peaches for canning by selecting firm, ripe fruit, washing them thoroughly, and peeling. You follow the exact recommended processing time using a boiling water canner, leaving the right headspace in sterilized jars. After processing, you cool the jars at room temperature without retightening the bands and store the sealed jars in a dark, cool place.

Why: This approach is safe because peaches are a high-acid fruit that can be safely processed in a boiling water canner. Using sterilized jars, following the correct processing time, allowing proper headspace, cooling jars correctly, and storing in a cool, dark place prevents growth of harmful microorganisms and spoilage.

Volunteer notes: High-acid fruits like peaches can be safely canned using boiling water canners with researched times and headspace. Jars must seal properly and not be retightened while hot to avoid seal failure. Proper cooling and storage maintains safety and quality. If a jar doesn't seal, it must be refrigerated and consumed soon or reprocessed within 24 hours, or the food destroyed. Follow PNW 199 Canning fruits for time and procedure.

Source: PNW 172 Canning vegetables - "When canning fruits, you can safely use a boiling water canner. Follow the preparation instructions carefully and process for the full time listed. Cool jars on a rack or cloth at room temperature, allowing air to circulate freely. Do not retighten screw bands after processing. Store jars in a cool, dark, dry place. For best eating quality and nutritive value, use within one year."

CF-T02 **NOT SAFE** *Teens (14-18)*

You filled jars with mixed fresh fruits and sealed them but didn't add any lemon juice or acid. You process the jars for less than 10 minutes in boiling water and did not adjust processing time for your altitude. You also overfilled the jars with fruit, leaving less than the recommended headspace.

Why: This is not safe because many fruits need added acid (like lemon juice) to ensure safe acidity levels for boiling water canning. Processing too short, not adjusting for altitude, or overfilling jars can prevent destroying harmful bacteria or achieving a proper vacuum seal. This may cause spoilage or botulism risk. Always follow tested recipes, add acid if called for, process full time, and maintain headspace.

Volunteer notes: Some fruits, especially those with lower acidity, require acid addition before boiling water canning to prevent growth of harmful bacteria. Inadequate processing time or overfilled jars cause underprocessing and unsafe food. Altitude affects boiling temperature and must be compensated for by longer processing. Follow PNW 199 and USDA guidelines strictly to avoid botulism and spoilage.

Source: PNW 172 Canning vegetables - "Some fruits...require the addition of acid (i.e., lemon juice) to be safely processed in a boiling water canner. Follow the preparation instructions carefully. Begin counting the processing time when the water comes back to a rolling boil. Adjust processing time for elevations over 1,000 feet. Leave 1/2 inch headspace for fruit and liquid to ensure good vacuum seals."

CF-T03 **IT DEPENDS** *Teens (14-18)*

You want to can slices of Asian pears at home. The recipe doesn't clearly say whether lemon juice is needed. You decide to can them in a boiling water canner, but you are unsure if you should add acid, process longer, or use a pressure canner instead.

Why: Safety depends on the acidity of the fruit and recommended process. Asian pears are borderline low-acid and may require adding lemon juice or other acid to safely process in a boiling water canner. If acid isn't added, pressure canning may be needed. Always follow research-based instructions for specific fruit types and recipes to ensure safety.

Volunteer notes: Some fruits like Asian pears and figs require acidification for safe boiling water canning. If acid is not added, they may need pressure canning. Decisions depend on acidity, processing method, and recipe. Incorrect method risks unsafe food or spoilage. Refer to PNW 199 and OSU/USDA tested recipes for exact directions.

Source: PNW 172 Canning vegetables - "Some fruits, such as Asian pears and figs, require the addition of acid (i.e., lemon juice) to be safely processed in a boiling water canner."

CF-A01 **SAFE** *Adults*

You have selected fresh, firm peaches and prepared them according to a tested recipe by washing, peeling, and slicing. You packed them into clean, hot Mason jars with the recommended headspace and processed them in a boiling water canner for the exact time specified in the recipe, adjusting for your altitude.

Why: Following a research-tested recipe for canning fruits ensures safety by using the correct processing time, temperature, and headspace. Boiling water canning is safe for high-acid fruits when volume, acidity, and jar size are correct, and altitude adjustments are made. Using clean, heat-tempered jars and new lids also ensures a proper seal and prevents spoilage.

Volunteer notes: Safety standards require using tested recipes with exact processing times and altitude adjustments for boiling water canning of fruits. Proper headspace and jar sterilization are essential for good seals and to prevent spoilage or pathogen growth.

Source: PNW 199 Canning fruits - "The directions in this publication have been carefully researched for safe home canning. Following the directions exactly is vital to ensure a product that is top quality and safe to eat... The canner must have room for 1 to 2 inches of briskly boiling water over the tops of the jars during processing... Standard Mason jars are the best choice for canning... Boil 10 minutes at altitudes less than 1,000 feet. Add 1 minute for each additional 1,000 feet of elevation."

CF-A02 **NOT SAFE** *Adults*

You decided to can underripe plums without adding any acid or adjusting the boiling water canner processing time. You packed the fruit raw into jars without preheating or hot packing and processed them for less time than recommended in your source recipe.

Why: Underripe fruit is less acidic and may not be safely processed with standard boiling water canning times. Raw packing without preheating and underprocessing can lead to spoilage or growth of dangerous microorganisms, including botulism. Always use ripe fruit and follow tested recipes and processing times precisely.

Volunteer notes: Fruits must be ripe and acidified if necessary to ensure safe acidity levels. Underprocessing, raw packing when not recommended, or skipping hot pack steps risks survival of spoilage or pathogenic microorganisms. Always adhere to tested guidelines for fruit ripeness and processing times.

Source: PNW 199 Canning fruits - "Acidity decreases as fruit ripens... Work quickly throughout preparation and canning. If food is allowed to stand, quality is lowered, and food spoilage is more likely to occur... Follow the directions exactly to ensure food safety."

CF-A03 **IT DEPENDS** *Adults*

You want to can pear halves at 3,000 feet altitude. The recipe you found shows times for sea level but lacks altitude processing time adjustments. You prepare the jars and pears as directed but are unsure if processing at sea level times is enough.

Why: Processing times for canning fruits must be adjusted for altitude as boiling temperature decreases at higher elevations. Not adjusting for altitude can result in inadequate heat treatment, risking spoilage or unsafe food. You must increase processing time or boiling water temperature by following altitude adjustment guidance to ensure safety.

Volunteer notes: Altitude affects boiling temperature, so processing times or pressure must increase at elevations above 1,000 feet. If a recipe lacks altitude instructions, contact local extension or trusted references for correct adjustment. This prevents underprocessing and botulism risk.

Source: PNW 199 Canning fruits - "Boil 10 minutes at altitudes less than 1,000 feet. Add 1 minute for each additional 1,000 feet of elevation... Following the directions exactly is vital to ensure a product that is top quality and safe to eat."

CANNING TOMATOES & SALSAS

CT-K01 **SAFE** *Kids (under 10)*

You have a jar of salsa that you made following a recipe from your cooking book. You used ripe tomatoes, bottled lemon juice, and cooked the salsa in a boiling water canner for the recommended time.

Why: This salsa is safe because you followed a tested recipe that has enough acid to stop bad germs from growing. Using bottled lemon juice and cooking it in a boiling water canner makes sure the salsa stays safe to eat.

Volunteer notes: Salsa recipes must have enough acidity to prevent botulism. Only use tested acidified recipes processed in a boiling water canner for safety. Follow exact times and ingredients to avoid risk.

Source: PNW 395 Salsa recipes for canning - "The salsa recipes in this publication have been tested under laboratory conditions to ensure that they contain enough acid to be processed safely in a boiling water canner."

CT-K02 **NOT SAFE** *Kids (under 10)*

You found a jar of canned tomatoes in the pantry that has a lid popped up and you smell a strange, bad odor coming from inside the jar.

Why: The popped lid and bad smell are signs that the tomatoes went bad. This can happen if the jar wasn't sealed right or if bad germs grew inside. Never eat canned food if the lid pops up, it smells bad, or looks funny.

Volunteer notes: Bulging lids, off-odor, leaking, or mold in home-canned tomatoes indicate spoilage and risk of botulism. Such jars should be discarded and not tasted to prevent foodborne illness.

Source: PNW 300 Canning tomatoes and tomato products - "Never taste food from a jar with an unsealed lid or food that shows signs of spoilage."

CT-K03 **IT DEPENDS** *Kids (under 10)*

You made salsa using a new recipe, but you weren't sure if the amount of vinegar was just right. After canning it, you wonder if it's safe to eat.

Why: Salsa safety depends on having enough acid like vinegar or lemon juice. If there's not enough acid, harmful germs can grow. To know if it's safe, you need to follow a tested recipe or keep the salsa in the fridge or freezer instead.

Volunteer notes: Salsa acidity is critical for safe boiling water canning. Insufficient acid or incorrect substitutions can cause botulism risk. Untested recipes should be refrigerated or frozen rather than canned.

Source: PNW 395 Salsa recipes for canning - "Because salsas are a mixture of acid and low-acid ingredients, they are an example of an acidified food appropriate for boiling water canning if-and only if-the level of acidity is adequate to prevent production of the botulism toxin."

CT-W01 **SAFE** *Tweens (10-14)*

You follow a tested recipe for tomato salsa, add the recommended amount of lemon juice, and process the jars in a boiling water canner for the full time. After canning, the jars seal tightly and you store them in a cool, dark place.

Why: When canning tomato salsa, it's important to use a recipe tested for safety that adds enough acid, like lemon juice, to stop bacteria from growing. Processing the jars for the right time in a boiling water canner kills harmful germs. Proper sealing and storage keeps your salsa safe to eat.

Volunteer notes: Tomato salsas are acidified foods; adding tested amounts of lemon juice or citric acid and following tested boiling water canning times prevents botulism. Unsafe recipes or processes increase risk. Acidified tomatoes/salsas are safe with proper processing and sealing.

Source: PNW 395 Salsa recipes for canning - "Salsa recipes for canning with a boiling water canner must meet acidity-level requirements to prevent the growth of botulism bacteria... follow directions carefully for each recipe in this publication."

CT-W02 **NOT SAFE** *Tweens (10-14)*

A family canned whole tomatoes that were very soft and from frost-killed vines, then packed them in jars without adding any lemon juice or citric acid before boiling water canning.

Why: Tomatoes from frost-killed or overripe fruit can lose acidity, making them unsafe to can without extra acid added. Without enough acid, bacteria like Clostridium botulinum can grow during storage and cause serious illness.

Volunteer notes: Frost-killed and overripe tomatoes lose acidity needed to safely process in boiling water canners. Adding lemon juice or citric acid is required for safe home canning of tomatoes. Using tomatoes without enough acid poses a botulism risk.

Source: PNW 300 Canning tomatoes and tomato products - "Over ripe tomatoes or tomatoes from frost-killed vines should not be used for canning because they lose acidity as they mature or if they freeze on the vine."

CT-W03 **IT DEPENDS** *Tweens (10-14)*

After canning tomato salsa, the family finds one jar didn't seal. They're wondering if it's still safe to eat if they keep it in the refrigerator and use it soon.

Why: If a jar doesn't seal, the canned salsa can spoil or let bacteria grow. But if you keep it cold in the fridge and use it quickly, it can be safe. Don't store unsealed jars at room temperature because harmful germs could grow.

Volunteer notes: Unsealed jars should not be stored on shelves. Refrigerate and consume quickly to prevent spoilage or risk of botulism. If left unrefrigerated, the product poses a serious health hazard.

Source: PNW 300 Canning tomatoes and tomato products - "If the suspect jars are unsealed, open, or leaking... Never taste food from a jar with an unsealed lid or food that shows signs of spoilage."

CT-T01 **SAFE** *Teens (14-18)*

Jamie used fresh, high-quality Roma tomatoes, peeled and cored them, then added bottled lemon juice before canning the salsa using a boiling water canner for the time recommended in a tested recipe.

Why: This salsa is safe because Jamie followed a tested recipe that ensures enough acidity by adding lemon juice to tomatoes. Using high-quality tomatoes and processing in a boiling water canner properly prevents harmful bacteria like botulism.

Volunteer notes: Tomatoes are borderline acid foods that require added acid (like bottled lemon juice) for safe boiling water canning. Using tested recipes and correct processing times prevents botulism. A typical follow-up question is why bottled lemon juice is needed instead of fresh.

Source: PNW 395 Salsa recipes for canning - "The salsa recipes in this publication have been tested under laboratory conditions to ensure that they contain enough acid to be processed safely in a boiling water canner... Use only high-quality tomatoes... Never can salsas that do not follow these or other research-tested recipes."

CT-T02 **NOT SAFE** *Teens (14-18)*

Alex canned homemade salsa but reduced the amount of vinegar in the recipe by half and added fresh garlic and fresh herbs before processing in a boiling water canner.

Why: This salsa is unsafe because reducing vinegar lowers acidity, which increases risk for botulism. Adding fresh garlic and herbs before canning also affects acidity and safety. Both changes can allow harmful bacteria to grow during boiling water canning.

Volunteer notes: Any reduction of acid (vinegar, lemon juice) makes the product unsafe for boiling water canning. Fresh garlic and herbs alter acidity and must not be added pre-canning. This leads to dangerous conditions for Clostridium botulinum growth.

Source: PNW 395 Salsa recipes for canning - "Never reduce the amount of vinegar, lemon juice, or lime juice in a recipe... It is not safe to add or increase the amounts of fresh herbs or garlic before canning because they affect the acidity level."

CT-T03 **IT DEPENDS** *Teens (14-18)*

Taylor wants to make a salsa with tomatoes and add extra corn and black beans before canning in a boiling water bath. They wonder if it will be safe to do so.

Why: Adding extra corn and beans before canning changes the acid balance and may lower the acidity, increasing botulism risk. These salsas must be canned according to tested recipes that account for such ingredients. If no tested recipe exists, it's safer to freeze instead.

Volunteer notes: Additional ingredients not listed in tested recipes, such as thickeners or other vegetables/beans, can make a salsa unsafe by lowering acidity. Canning safety depends on acidity level and recipe testing. Freezing is recommended for untested recipes.

Source: PNW 395 Salsa recipes for canning - "Adding ingredients not listed in a salsa recipe will result in an unsafe product if done before canning... Salsa can be thickened after opening... The flavor can be enhanced with additional ingredients such as corn, black beans, or other additions just before serving."

CT-A01 **SAFE** *Adults*

You followed a tested salsa recipe that included adding bottled lemon juice to your tomatoes and onions, processed the jars in a boiling water canner for the full 15 minutes at your local altitude, and used fresh, firm Roma tomatoes.

Why: This salsa is safe because it follows a research-tested recipe that ensures enough acidity by adding bottled lemon juice, uses the proper processing time in a boiling water canner, and uses high-quality, fresh tomatoes. Processing time and acidity are critical to prevent botulism in salsas, which contain low-acid ingredients mixed with tomatoes. Following exact recipes and methods that have been lab-tested guarantees safety.

Volunteer notes: Tomato-based salsas must have tested acidity levels often ensured by adding bottled lemon or lime juice, and must be processed for the exact time recommended in a boiling water canner. Use only tested recipes to ensure safety .

Source: PNW 395 Salsa recipes for canning - "Salsa recipes have been tested under laboratory conditions to ensure enough acid for safe boiling water canning."

CT-A02 **NOT SAFE** *Adults*

You canned salsa using fresh tomatoes but did not add any bottled lemon juice or vinegar, and processed the jars in a boiling water canner for only 5 minutes instead of the recommended 15.

Why: Not adding bottled lemon juice or vinegar can result in inadequate acidity, allowing Clostridium botulinum bacteria to grow, which can cause botulism-a serious foodborne illness. Also, under-processing (shorter than recommended time) does not safely eliminate bacteria. Salsa must have controlled acidity and proper processing time to be safe.

Volunteer notes: Acidity and processing time are critical for safety in canning acidified foods like salsa. Omitting acid or reducing processing time below tested levels risks botulism toxin production .

Source: PNW 395 Salsa recipes for canning - "If the mixture has less acidity, it needs to be treated as a low-acid food ... Never can salsas that do not follow research-tested recipes."

CT-A03 **IT DEPENDS** *Adults*

You made salsa with fresh garden tomatoes and peppers using your own original recipe that does not specify vinegar or bottled lemon juice amounts, then processed it in a boiling water canner for 15 minutes at sea level.

Why: The safety depends on the exact acidity level of your salsa. Without a tested recipe specifying acid type and amount, you cannot be sure the acid level is sufficient to prevent botulism. The processing time is adequate if acidity is correct, but unknown acidity means the risk cannot be ruled out. Untested recipes should only be frozen or refrigerated,

not canned.
Volunteer notes: Original salsa recipes without tested acid levels are unsafe to can because inadequate acidity allows bacterial toxin growth. Acidified foods require strict acid and processing controls; untested recipes are only safe frozen or refrigerated .
Source: PNW 395 Salsa recipes for canning - "It's not safe to can your own original salsa recipe. Refrigerate or freeze it instead."

LOW ACID FOODS

LA-K01 SAFE *Kids (under 10)*

Mom packed cooked green beans into jars and processed them in a pressure canner for the right time. The lid sealed tight, and the jars are cool now. Can we eat them safely from the jar?

Why: Because green beans are low acid, they must be processed in a pressure canner to reach a high enough temperature that kills harmful bacteria. When canned this way and sealed tight, the food is safe to eat. Pressure canning protects us from dangerous germs like botulism that can grow in low acid foods if not heated enough.
Volunteer notes: Low acid foods such as vegetables must be pressure canned to reach 240°F to kill Clostridium botulinum spores. Proper sealing and processing time are critical for safety. Low acid foods cannot be safely canned in boiling water canners.
Source: PNW 361 Canning meat poultry game - "Because meat and poultry products are low-acid foods, Clostridium botulinum can survive and grow in these foods when sealed in oxygen free jars. Only pressure canning produces temperatures high enough (240°F) to kill these bacteria."

LA-K02 NOT SAFE *Kids (under 10)*

You find a jar of canned green beans in the pantry, but the lid popped up like a little mountain before you opened it. Is it safe to eat?

Why: When the lid pops up, it means the jar did not seal properly and bad germs could have grown inside. These germs can make the food unsafe even if it looks or smells okay. It's best to throw the jar away to stay safe.
Volunteer notes: A lid that bulges or pops up indicates seal failure and potential bacterial growth including dangerous botulinum toxin. Unsafe low acid canned foods with unsealed lids must be discarded to prevent botulism poisoning.
Source: PNW 361 Canning meat poultry game - "If meat products are improperly processed, the botulism toxin could be present even though the canned food looks, smells, and tastes normal."

LA-K03 IT DEPENDS *Kids (under 10)*

A jar of cooked carrots was left out on the counter all day after canning. The lid sealed tight, but it smells a little funny. Is it safe to eat?

Why: Low acid foods need to be kept hot or refrigerated after opening because bacteria can grow if left out too long. Even if the jar sealed initially, if it smells funny or shows mold, it is not safe. But if it smells fresh and has no mold, cooking it again can make it safe.
Volunteer notes: Low acid foods are susceptible to bacterial growth if held too long at unsafe temperatures after processing or opening. Sensory signs (bad smell, mold) indicate spoilage. Reheating can destroy some bacteria but not toxins. When in doubt, throw it out.
Source: PNW 450 Canning smoked fish at home - "Low acid foods (especially cooked ones) that have been kept in the danger zone (40-140°F) for more than 2-3 hours are a safety concern. Watch for smell, mold, or bubbling signs."

LA-W01 SAFE *Tweens (10-14)*

Jamie canned fresh green beans in jars using a pressure canner following the exact time and pressure instructions from a trusted recipe. The jars sealed properly and were stored in a cool, dark pantry.

Why: Green beans are low-acid foods, so they must be canned in a pressure canner to reach high enough temperatures to kill harmful bacteria like Clostridium botulinum. Using tested recipes and following exact pressure and time ensures safety. Properly sealed jars stored in a cool, dark place remain safe to eat.
Volunteer notes: Low acid foods require pressure canning at 240°F to destroy botulism spores. Follow tested recipes precisely. A boiling water canner or incorrect times are unsafe for low acid foods. Proper jar sealing and storage are critical to prevent spoilage.
Source: PNW 361 Canning meat poultry game - "Only pressure canning produces temperatures high enough (240°F) to kill bacteria which grow in low-acid foods, including Clostridium botulinum. Follow directions exactly for filling jars and processing times."

LA-W02 **NOT SAFE** *Tweens (10-14)*

Alex canned leftover cooked meat in jars but used a boiling water bath canner instead of a pressure canner. The jars sealed and look fine, but Alex wants to eat the meat soon.

Why: Cooked meat is a low-acid food that can harbor botulism bacteria unless canned at high temperature in a pressure canner. Boiling water bath canners don't reach a high enough temperature, so contamination and toxin risk remain, even if jars appear sealed and food looks normal.

Volunteer notes: Only pressure canning is safe for meat or other low acid foods. Boiling water bath canners cannot kill botulinum spores. Eating food canned improperly can cause deadly botulism poisoning. Warn to discard or reprocess properly.

Source: PNW 361 Canning meat poultry game - "Because meat products are low-acid foods, Clostridium botulinum can survive and grow when sealed in jars. Only pressure canning produces necessary temperatures to kill these bacteria."

LA-W03 **IT DEPENDS** *Tweens (10-14)*

Morgan made homemade garlic oil and stored it in the fridge, but wonders if it's safe to keep it for up to a month. The oil covers the garlic completely and it's refrigerated constantly.

Why: Garlic in oil is a low-acid food that can grow botulism bacteria if not stored properly. Keeping it refrigerated helps, but homemade garlic oils are safe for only about 3 weeks unless they're acidified. After that, the risk of botulism toxin increases, so safety depends on how long and how cold it's stored.

Volunteer notes: Homemade garlic in oil can support botulism growth at room temperature; refrigeration slows this but safety is limited to ~3 weeks unless acidified. Commercial garlic oils are acidified for safety. Always label and date homemade garlic oils.

Source: PNW 450 Canning smoked fish at home - "Long term refrigerated storage of garlic or herbs in oil is associated with a risk of botulism. Homemade garlic or herbs in oil refrigerated; use within 3 weeks."

LA-T01 **SAFE** *Teens (14-18)*

You want to can green beans, which are low-acid vegetables. You follow a tested recipe precisely, use a pressure canner, maintain the correct pressure and time, and check your dial gauge annually.

Why: Low-acid foods like green beans must be pressure canned to kill bacteria like Clostridium botulinum. Using a tested recipe with the correct pressure, time, and equipment ensures safety. Checking your canner's dial gauge each year ensures it works properly.

Volunteer notes: Low acid foods (pH above 4.6) require pressure canning at 240°F to eliminate botulism spores. Following USDA recipes and equipment checks prevents dangerous toxins. Improper processing can cause deadly botulism toxin in sealed jars.

Source: PNW 361 Canning meat poultry game - "Poultry, red meats, and meat from game animals ... are low-acid foods and must be processed in a pressure canner ... to kill botulism-causing bacteria ... each time you use a pressure canner, check the petcock or vent port ... Follow directions exactly for filling jars ... Never can meat products or combination foods for which you do not have research-based processing times."

LA-T02 **NOT SAFE** *Teens (14-18)*

You canned cooked potatoes in a boiling water bath canner instead of a pressure canner and stored the jars at room temperature.

Why: Cooked potatoes are low-acid foods and need pressure canning to reach temperatures high enough to kill botulism spores. Using only boiling water canning is not safe and can allow harmful bacteria to grow, even if the jar looks fine.

Volunteer notes: Boiling water canners do not reach 240°F needed for low acid foods; using them for potatoes risks botulism poisoning. Low acid foods require pressure canning; improper canning is a leading cause of botulism. Always use pressure canner for meats, vegetables, and potatoes.

Source: PNW 361 Canning meat poultry game - "Only pressure canning produces temperatures high enough (240°F) to kill many bacteria which can grow in low-acid foods, including Clostridium botulinum ... Boiling water canners and steam canners do not produce temperatures high enough to kill botulism-causing bacteria."

LA-T03 **IT DEPENDS** *Teens (14-18)*

You canned a mixture of meat and vegetables in a pressure canner but shortened the recommended processing time by 5 minutes to save time.

Why: Low acid foods need precise processing times to destroy dangerous bacteria. Shortening the time may result in

unsafe food. Whether it's safe depends on how much time was cut, the type of food, and the canner's temperature. It's safer to follow tested times exactly.
Volunteer notes: Processing time and pressure must meet tested USDA standards for low acid foods; deviations can let botulism spores survive. Even small time reductions can make the food unsafe. Always follow recommended times and pressure for the exact food and altitude.
Source: PNW 361 Canning meat poultry game - "Follow directions exactly for filling jars ... Never can meat products or combination foods for which you do not have research-based processing times. A safe canning time cannot accurately be determined at home."

LA-A01 **SAFE** *Adults*

Sarah uses a tested USDA recipe to pressure can green beans in quart jars. She maintains correct headspace, adjusts processing time for her 3,000 feet altitude, and ensures her pressure canner's gauge is accurate before starting.

Why: Using a tested recipe with the correct processing time, headspace, and altitude adjustment ensures the destruction of Clostridium botulinum spores in low-acid foods. Correct equipment maintenance, like accurate pressure gauges, is critical for safety. Pressure canning reaches the necessary temperature (240°F) that boiling water canners cannot achieve.
Volunteer notes: Low acid foods must be pressure canned using tested recipes that specify time, pressure, and altitude adjustments to prevent botulism. Use a properly maintained pressure canner with accurate gauges. Always follow USDA or OSU Extension recommendations exactly.
Source: PNW 361 Canning meat poultry game - "Only pressure canning produces temperatures high enough (240°F, 28 degrees above boiling) to kill many bacteria which can grow in low-acid foods, including Clostridium botulinum... Increase pressure at altitudes above 1,000 feet for weighted-gauge canners or 2,000 feet for dial-gauge canners to reach the proper temperature (240°F) for pressure canning."

LA-A02 **NOT SAFE** *Adults*

Tom fills quart jars with leftover spaghetti sauce with meat, puts on lids, and processes them in a boiling water canner for 20 minutes, then stores them at room temperature without refrigeration.

Why: Spaghetti sauce with meat is a low-acid food and must be processed in a pressure canner, not a boiling water canner. Boiling water cannot reach the 240°F needed to kill botulism spores. Processing low-acid foods incorrectly risks deadly botulism poisoning, even if the food looks and smells normal.
Volunteer notes: Low-acid foods like meat sauces must be pressure canned to kill botulism spores. Boiling water canners are only safe for high-acid foods. Improper processing can produce a fatal toxin that cannot be detected by smell or appearance.
Source: PNW 361 Canning meat poultry game - "Because meat and poultry products are low-acid foods, Clostridium botulinum ... can survive ... Only pressure canning produces temperatures high enough (240°F)... Boiling water canners and steam canners do not produce temperatures high enough to kill botulism-causing bacteria."

LA-A03 **IT DEPENDS** *Adults*

Linda stores homemade garlic-in-oil mixtures at room temperature, but she only followed some acidification steps and is unsure if the product is sufficiently acidified to prevent botulism.

Why: Garlic in oil is a low-acid food and can support Clostridium botulinum growth without proper acidification. Commercial products are acidified to prevent bacterial growth. Safety depends precisely on following tested acidification procedures; if acidification is incomplete or inconsistent, the product is unsafe at room temperature. Otherwise, it must be refrigerated or frozen.
Volunteer notes: Low-acid foods in oil must be acidified per tested methods to prevent botulism growth. If unsure about acidification or recipe adherence, refrigeration or freezing is required. Following research-based guidelines like those in PNW 664 is critical.
Source: PNW 450 Canning smoked fish at home - "Low acid foods stored in oil creates an oxygen-free environment which is perfect for growth of Clostridium botulinum bacteria... Commercial garlic-in-oil mixtures are acidified to prevent bacterial growth... Research has been done on acidification procedures... The acidification directions must be followed exactly."

PICKLING

PK-K01 **SAFE** *Kids (under 10)*

You are making quick pickles by adding vinegar to your cucumbers using a clean jar and fresh ingredients. You use regular vinegar from the store with the right sour taste and keep the jar closed in the fridge.

Why: When you use real vinegar with enough sourness and keep everything clean, the pickles stay safe to eat. The

vinegar stops bad germs from growing. It's important to not use homemade vinegar and to use fresh vegetables and a clean jar.
Volunteer notes: Quick pickles must use vinegar with 5% acidity to prevent harmful bacteria growth. Using fresh, unblemished vegetables, clean jars, and following tested recipes keeps pickles safe in the fridge.
Source: PNW 355 Pickling vegetables - "Vinegar is the most important ingredient in quick-pickle recipes. Use any vinegar with 5% acidity... Do not use homemade vinegar. Use fresh vegetables and clean containers."

PK-K02 NOT SAFE *Kids (under 10)*

You find a jar of homemade pickles on the counter that was left open for a whole day, and the lid is popped up. The pickles look cloudy and smell funny.

Why: If the jar lid is popped up and the pickles smell bad or look moldy or cloudy, this means bad bacteria or germs have grown. It is unsafe to eat, and you should throw it away to not get sick.
Volunteer notes: Popped lids, bad smells, mold, or cloudiness are signs of spoilage and possible harmful bacteria like Clostridium botulinum. Unsafe pickles can cause foodborne illness and must be discarded.
Source: UC David Safely fermented fruit and vegetable - "Botulism can occur if fermentation is not successful and acid levels are low... Watch for visible signs that the food might be spoiled such as mold, bubbles rising, and discoloration."

PK-K03 IT DEPENDS *Kids (under 10)*

Your pickles have been sitting on the counter at room temperature for a few days without being covered tightly. You notice a slight bubbly sound and the jar lid moved just a little.

Why: Sometimes bubbling means the pickles are fermenting safely, making good acid to protect them. But if the jar wasn't clean, or the lid popped open a lot, or the smell is bad, it might be dangerous. Check if it smells nice and sour and has no mold. If it smells bad or looks fuzzy, it's not safe.
Volunteer notes: Fermentation produces carbon dioxide which can cause bubbling and slight lid movement; this is normal if acid levels are sufficient. However, safety depends on proper sanitation, acidity, and no signs of spoilage like bad odor or mold.
Source: UC David Safely fermented fruit and vegetable - "Gas produced by fermentation may cause breakage... Proper sanitation and acid production inhibit growth of harmful bacteria. When in doubt, throw it out."

PK-W01 SAFE *Tweens (10-14)*

You want to make quick pickles by adding vinegar to sliced cucumbers. You use white distilled vinegar with 5% acidity and follow the recipe without changing the amount of vinegar. You keep the pickles refrigerated after sealing the jar.

Why: Using vinegar with 5% acidity is very important for safety because it prevents harmful bacteria from growing. Sticking closely to the recipe's vinegar amount ensures the pickles will be safe to eat. Keeping pickles refrigerated after sealing helps stop bacteria from growing.
Volunteer notes: Use vinegar with 5% acidity as required in tested pickling recipes to ensure acid levels prevent harmful bacteria growth. Follow OSU and USDA guidelines carefully for vinegar amounts and refrigeration to keep pickles safe.
Source: PNW 355 Pickling vegetables - "Vinegar is the most important ingredient in quick-pickle recipes. Use any vinegar with 5% acidity. Do not reduce the amount of vinegar or increase water. Keep pickles refrigerated."

PK-W02 NOT SAFE *Tweens (10-14)*

Someone soaked cucumbers in lime purchased from a garden center to make pickles. They forgot to rinse the cucumbers after soaking and then packed them in jars to pickle.

Why: Lime used for pickling must be food-grade and not garden or lumberyard lime, which can be toxic. Also, not rinsing cucumbers after soaking in lime can leave harmful chemicals that are unsafe to eat.
Volunteer notes: Only use food-grade lime for soaking in pickling, and be sure to rinse cucumbers thoroughly after soaking to remove excess lime. Garden center lime is not safe and can cause poisoning if used.
Source: PNW 355 Pickling vegetables - "Do not use lime sold at garden centers or lumberyards. Excess lime must be removed by rinsing cucumbers thoroughly before pickling."

PK-W03 IT DEPENDS *Tweens (10-14)*

After making fermented pickles in a crock, you left the vegetables uncovered at room temperature for a week. Now there are some bubbles and a bit of mold on the surface.

Why: Fermented pickles need to be kept under the brine and covered to prevent mold and insects. Some bubbles are normal from fermentation, but mold can be unsafe. Safety depends on whether the mold is just on the surface and if you

remove it or if the fermentation was done correctly.
Volunteer notes: During fermentation, vegetables must stay submerged under brine and container covered with a cloth to prevent mold and contamination. Bubbles usually mean good fermentation. Surface mold might be removed safely under guidance, but widespread mold means discard product.
Source: PNW 355 Pickling vegetables - "Cabbage and cucumbers must be kept 1 to 2 inches under the brine while fermenting. Cover the container opening to prevent contamination from insects and molds."

PK-T01 SAFE *Teens (14-18)*

A teenager follows a tested recipe for refrigerator pickles using vinegar with 5% acidity, fresh cucumbers, and sterilized jars. They keep the pickles refrigerated and consume them within 2 weeks.

Why: Using a tested recipe with vinegar that has 5% acidity ensures the pickles are acidic enough to prevent harmful bacteria. Refrigeration slows down spoilage. Consuming within 2 weeks means the pickles remain fresh and safe.
Volunteer notes: Vinegar with 5% acidity is required for safe pickling because it keeps the food's pH below 4.6, which prevents growth of botulism bacteria. Refrigeration is necessary for quick refrigerated pickles. Follow tested recipes to ensure safety.
Source: NCHFP Pickled eggs - "Use vinegar with 5% acidity... Vinegar is important for safety reasons... Keep pickles refrigerated and consume within 2 weeks ."

PK-T02 NOT SAFE *Teens (14-18)*

Someone tries to make pickles using homemade vinegar without knowing its acidity level and skips processing the jars in boiling water. They store the jars at room temperature for months.

Why: Homemade vinegar with unknown acidity may not be acidic enough to prevent harmful bacteria like Clostridium botulinum. Skipping boiling water processing risks unsafe contamination. Storing at room temperature increases the danger of bacteria growth.
Volunteer notes: Only vinegar with a known 5% acidity should be used for pickling to ensure the acid level is high enough to prevent botulism. Proper heat processing (like boiling water bath for high-acid foods) is essential. Lack of these steps creates a food safety risk.
Source: NCHFP Pickled eggs - "Do not use homemade vinegar or brands with an unknown acidity. The percent acidity must be known for successful pickling... Fruits and pickles (high acid foods) can be safely processed in a boiling water canner ."

PK-T03 IT DEPENDS *Teens (14-18)*

A teen is making fermented pickles by following a traditional fermentation method but wonders if they have to add enough salt and keep the fermentation environment clean to ensure safety.

Why: Fermentation safety depends on using the proper amount of salt to encourage good bacteria and keep bad bacteria out. Clean equipment and fresh vegetables are also needed. If these conditions aren't met, harmful bacteria might grow.
Volunteer notes: Successful fermentation requires proper sanitation, using fresh ingredients, and correct salt concentrations to produce enough acid to inhibit pathogens like botulism. If these are uncertain or not followed, the product may not be safe.
Source: UC David Safely fermented fruit and vegetable - "Proper sanitation, high quality ingredients, and following researched recipes are key to fermentation success... The production of sufficient acid will inhibit growth of Clostridium botulinum ."

PK-A01 SAFE *Adults*

A home cook follows a tested pickling recipe exactly, using 5% vinegar, packing vegetables in clean Mason jars with proper headspace, and processing in a boiling water canner for the recommended time based on altitude.

Why: Following a tested recipe ensures the correct acidity and processing time to prevent bacterial growth and spoilage. Using the proper vinegar concentration and processing time adjusted for altitude guarantees safety. Proper headspace and clean jars prevent contamination and allow for adequate heat penetration during processing.
Volunteer notes: OSU/USDA standards require using tested recipes with verified vinegar acidity (5%), correct headspace (usually 1/2 inch), and boiling water bath processing adjusted for altitude to ensure destroying harmful bacteria like Clostridium botulinum. Common questions include how altitude affects processing times.
Source: PNW 355 Pickling vegetables - "Use any vinegar with 5% acidity. Caution: Do not reduce the amount of vinegar or increase the amount of water in a recipe. Processing times vary by altitude; adjust accordingly for safety"

PK-A02 **NOT SAFE** *Adults*

Someone uses homemade vinegar of uncertain acidity to make pickles and does not adjust the recipe's vinegar amount or processing time. The jars are sealed without being processed in boiling water or a pressure canner.

Why: Homemade vinegar can vary widely in acidity and may not reach the critical 5% acidity needed to safely inhibit bacterial growth. Without proper acid levels and heat processing, harmful bacteria, including Clostridium botulinum, can grow leading to poisoning. Processing ensures destruction of pathogens.

Volunteer notes: Using untested vinegar acidity and skipping heat processing violates OSU/USDA safe canning standards and risks botulism. Vinegar must be 5% acidity at minimum, and jars require boiling water bath processing for quick pickles. Expect questions about how to test vinegar acidity safely.

Source: PNW 355 Pickling vegetables - "Do not use homemade vinegar. It varies in acidity. Use vinegar with 5% acidity. Processing times are critical to prevent bacterial growth and spoilage"

PK-A03 **IT DEPENDS** *Adults*

A home preserver intends to make quick pickles but is unsure whether they should boil the jars in a water bath after packing with vinegar and vegetables stored at 4,000 feet elevation.

Why: Quick pickles require boiling water processing to ensure safety, but processing time must be adjusted for altitude. At 4,000 feet, processing time is longer to compensate for lower boiling temperature. Without proper adjustment, safety is not guaranteed.

Volunteer notes: OSU/USDA guidelines specify longer boiling water bath times with increased altitude for safety. A typical pint jar might need 40 minutes at 3,001-6,000 feet instead of 30 minutes at sea level. Confirm altitude and exact recipe instructions. Common FAQs include altitude adjustments.

Source: PNW 355 Pickling vegetables - "Processing times for quick pickled vegetables increase with altitude: e.g., quart jars processed 30 min at 0-1,000 ft, 40 min at 3,001-6,000 ft, and more above 6,000 ft"

JAMS & JELLIES

JJ-K01 **SAFE** *Kids (under 10)*

Your mom made strawberry jam and boiled the jars in hot water for the right time. The jar lid is popped down tight, and the jam smells sweet and fresh. You can see no fuzzy stuff or bubbles in the jar.

Why: When jam jars are boiled for the proper time, sealed tightly, and look clear without mold or bubbles, they are safe to eat. The popping lid shows a good vacuum seal that keeps air and germs out. Always check the smell and look to make sure nothing is wrong before eating.

Volunteer notes: Proper processing in a boiling water canner creates a vacuum seal and destroys mold spores on jams and jellies. Signs of a good seal include a lid popped down tight, no movement when pressed, and clear appearance without mold or bubbles. Use sensory checks before eating.

Source: SP 50-632 Making berry syrups at home - "Testing for Seal After cooling 12-24 hours, test each jar for a vacuum seal. The lid has popped down in the center. ... Look for any unnatural color, bubbles, or cotton-like mold growth. Smell for off odors."

JJ-K02 **NOT SAFE** *Kids (under 10)*

You open a jar of your sweet grape jelly and see fuzzy green spots on top. The jelly smells weird and sour, not like the yummy jelly you remember. The lid was popped up when you tried to press it.

Why: If you see fuzzy mold or green spots on jelly, or the jelly smells bad or sour, it means it can be dangerous to eat. Also, if the lid is popped up, it shows no seal, and germs can spoil the jelly. Always throw away jelly with mold or off smells.

Volunteer notes: Mold on jams and jellies indicate spoilage and possible bacterial growth. Surface mold and off odors mean the product is unsafe. No vacuum seal (lid popped up) raises risk of contamination. Such products should be discarded.

Source: SP 50-632 Making berry syrups at home - "Foods with high moisture content can be contaminated below the surface. Moldy foods may also have bacteria growing along with the mold. The mold could be producing a mycotoxin... The microbiologists recommend against using remaining jelly."

JJ-K03 **IT DEPENDS** *Kids (under 10)*

Your grandma left homemade peach jam out on the kitchen counter all day. When you check, the jam looks normal and smells okay, but the lid is not popped down tight. You wonder if it is safe to eat.

Why: When jam is left out too long, it can grow germs, especially if the lid seal is broken. But if it looks clear, smells sweet and fresh, and there is no fuzzy mold, it might still be okay for a short time. Use your nose and eyes to decide, but when in doubt, don't eat it.

Volunteer notes: Jam is acidic, which helps prevent bacteria, but jams with broken seals left at room temperature too long can spoil. Absence of mold and off odors help judge safety, but extended time out increases risk of spoilage. Refrigerate after opening.

Source: SP 50-632 Making berry syrups at home - "Don't leave any perishables out of the refrigerator more than 2 hours... Use leftovers within 3 to 4 days so mold doesn't have a chance to grow. When you see moldy food, discard it."

JJ-W01 **SAFE** *Tweens (10-14)*

Emma followed a tested recipe to make strawberry jam, added the right amount of sugar and lemon juice, boiled it for the full time, then sealed the jars in a boiling water canner. She stored the jam jars in a cool, dark cupboard.

Why: Emma used a proven recipe with enough sugar and acid, boiled the jam to kill harmful germs, sealed the jars tightly, and stored them properly. This keeps the jam safe to eat for a long time.

Volunteer notes: Using tested recipes, processing times, and proper sealing in a boiling water canner ensures jams are safely preserved by killing spoilage and harmful bacteria in high-acid foods. Store in cool, dark places for best quality and safety.

Source: SP 50-765 Low sugar fruit spreads - "Follow the preparation instructions in the pectin package. Process low - and no - sugar spreads in a boiling water canner for 10 minutes. After the product is opened, it should be stored in the refrigerator."

JJ-W02 **NOT SAFE** *Tweens (10-14)*

Liam made jam but used less lemon juice than the recipe called for and didn't heat the jam long enough. He canned it and stored it at room temperature hoping it would last.

Why: Liam didn't add enough acid or cook the jam long enough, so it might not be acidic enough to stop bad bacteria from growing. This can make the jam spoil or even cause food poisoning.

Volunteer notes: Insufficient acidity (pH above 4.6) and under-processing can allow harmful bacteria like Clostridium botulinum to survive. Always follow tested recipes precisely, including acid additions and heating times.

Source: SP 50-765 Low sugar fruit spreads - "Simply reducing the sugar in a traditional jam or jelly recipe will result in a runny and potentially unsafe product. ... acidity level is high enough to prevent bacterial growth."

JJ-W03 **IT DEPENDS** *Tweens (10-14)*

Sophia made freezer jam using fresh raspberries and less sugar than usual. She stored the jars in the freezer but left one jar in the fridge for a week instead.

Why: Freezer jam is safe if kept frozen, since freezing stops bacteria. But if kept in the fridge for a week, it might spoil because it's not cooked like boiled jam. It's important to follow the recipe's storage instructions.

Volunteer notes: Uncooked freezer jams rely on freezing for safety and should be kept frozen until eaten. Refrigerated freezer jam can spoil quickly since it lacks preservatives from heat processing. Follow storage instructions precisely.

Source: SP 50-763 Uncooked freezer jams - "Spreads produced using these instructions are safe to process in jars in a water bath canner and store at room temperature. ... low - or no - sugar needed pectin"

JJ-T01 **SAFE** *Teens (14-18)*

You made a strawberry freezer jam using a tested recipe from a low-sugar pectin package and stored it in the freezer right after sealing the jars. You keep the freezer jam frozen until you use it.

Why: Freezer jams that use tested low-sugar pectin recipes and are kept frozen are safe. The freezing stops bacterial growth, and the tested recipe ensures the right acidity and pectin for safety and texture. Just follow the instructions on the pectin package and store the jam in the freezer until use.

Volunteer notes: Uncooked freezer jams made with low-sugar pectin must be stored frozen and consumed within a limited time to be safe. The pectin instructions tested for adequate acid and gel formation are crucial. Follow water bath canning rules for heated jams, but freezer jams rely on freezing to prevent spoilage and pathogens.

Source: SP 50-763 Uncooked freezer jams - "Uncooked freezer jams use special pectin and must be kept frozen to prevent bacterial growth and ensure safety. Follow tested recipes and instructions for freezing."

JJ-T02 **NOT SAFE** *Teens (14-18)*

You canned a batch of peach jam but skipped the boiling water bath processing step and just let the hot jam cool in jars. You plan to store it on your pantry shelf for several months.

Why: Skipping the boiling water bath means the jam isn't processed to kill bacteria and molds that can grow in home-canned food. Even acid jams need processing to be safe on the shelf. Without processing, your jam could spoil or cause food poisoning like botulism.

Volunteer notes: Home-canned jams and jellies must be processed in a boiling water bath for the recommended time to destroy molds and bacteria, even if the product is high acid. Otherwise, the product is unsafe for shelf storage. Freezer storage is an alternative not requiring processing.

Source: SP 50-764 Making jam jellies fruit spreads - "Jams and jellies need boiling water bath processing to be safe on the shelf and prevent mold and bacterial growth."

JJ-T03 **IT DEPENDS** *Teens (14-18)*

You want to make a jelly using elderberries you picked yourself. You know there are different varieties with different acidity levels, but you're not sure if yours is safe for standard canning.

Why: Elderberry jelly safety depends on the variety of elderberry used because some have too low acidity for safe home canning with regular recipes. The native blue elderberry is safe with normal boiling water canning, but European and American elderberries need special high sugar recipes or freezing to be safe. Always identify your berry variety and follow the correct recipe.

Volunteer notes: Different varieties of elderberries vary in acidity and cyanogenic compounds. Native blue elderberries are safe with standard recipes; others require modified recipes or freezing. Using the wrong recipe risks foodborne illness due to unsafe pH.

Source: SP 50-536 Wild berries and fruits - "Native Blue Elderberry is safe for boiling water canning; European and American varieties have pH too high for standard recipes and need special handling."

JJ-A01 **SAFE** *Adults*

You follow a tested low-sugar jam recipe that uses low-methoxyl pectin. After cooking, you process the jars in a boiling water bath canner for 10 minutes. You refrigerate the jam after opening.

Why: Using low-methoxyl pectin for low-sugar jams is safe because it's designed to set without high sugar levels. Processing in a boiling water canner for the recommended time ensures destruction of molds and pathogens. Refrigeration after opening prevents spoilage.

Volunteer notes: Low-sugar jams must use tested recipes with low-methoxyl pectin and be processed by boiling water bath for at least 10 minutes. Refrigerate after opening to inhibit microbial growth. This matches OSU/USDA safety guidelines for reduced sugar products.

Source: SP 50-632 Making berry syrups at home - "Spreads produced using these instructions are safe to process in jars in a water bath canner and store at room temperature. After the product is opened, it should be stored in the refrigerator."

JJ-A02 **NOT SAFE** *Adults*

You opened a jar of jelly stored at room temperature and noticed a fuzzy mold growth on the surface. You carefully scraped off the mold and planned to eat the jelly as usual.

Why: Mold on jams and jellies can contaminate the entire product below the surface, and molds may produce dangerous mycotoxins. It is unsafe to simply remove mold and consume the rest. The entire jar should be discarded to prevent illness.

Volunteer notes: Microscopic molds can grow in high-acid foods like jams, but they can penetrate beneath the surface and produce toxins. The USDA recommends discarding moldy jams and jellies rather than trying to salvage them by scraping mold off.

Source: SP 50-632 Making berry syrups at home - "Mold generally cannot penetrate deep into the product. ... Microbiologists recommend against scooping out the mold and using the remaining condiment."

JJ-A03 **IT DEPENDS** *Adults*

You made a pepper jelly using a recipe without specifying added acid. You processed the jelly in a boiling water bath canner for the usual recommended time. You wonder if it's safe to store at room temperature.

Why: Pepper jellies with low-acid vegetables depend on added acid (like vinegar or lemon juice) for safety. Without sufficient acidification, processing alone cannot guarantee safe acidity (pH below 4.6), risking botulism. Safety depends on following a tested recipe that includes acid.

Volunteer notes: Low-acid ingredients in jams/jellies (e.g., peppers) must have added acid and use tested recipes. Processing time alone is insufficient for safety without achieving pH under 4.6. Confirm acid levels or use tested recipes to ensure safety before room temp storage.

Source: SP 50-632 Making berry syrups at home - "Canned low sugar spreads made with low acid ingredients (like peppers and herbs) rely on added acid for safety. Do not reduce these acids."

FREEZING & COLD STORAGE

FZ-K01 **SAFE** *Kids (under 10)*

You put your blueberries in the freezer right after picking them and keep the freezer cold all the time. When you want a snack, you take some out still frozen. Is this safe?

Why: When frozen food is kept at the right cold temperature and stays frozen, it stops germs from growing. The blueberries stay safe to eat if they never thaw before you take them out. Keeping the freezer cold and not opening it too much helps keep the food good and safe.

Volunteer notes: Frozen foods stored at 0°F or below maintain safety by stopping microbial growth. It is safe to eat food directly from the freezer if it has remained frozen. Quality may last 8-12 months for fruits, but longer storage affects quality, not safety.

Source: PNW 214 Freezing fruits and vegetables - "Store them at 0°F (-18°C) or below. Foods lose quality ... but they will be safe to eat."

FZ-K02 **NOT SAFE** *Kids (under 10)*

Your friend left a bottle of canned soup in the unheated garage all winter and it froze. When she brought it inside, the can looks swollen and the lid is bulging. Is it safe to eat?

Why: When canned food freezes and the can lid bulges or seams break, it means germs might have grown inside and made the food bad. So it's best to throw it away to keep from getting sick because bulging cans mean danger.

Volunteer notes: If a can's lid bulges or seam is broken after freezing and thawing to room temperature, the food should be discarded due to possible spoilage or botulism risk. Safe practice is to discard damaged cans.

Source: PNW 296 Freezing convenience foods that you've prepared at home - "If the seam has broken and/or the lids are bulging and the food has thawed to room temperature, it should be discarded."

FZ-K03 **IT DEPENDS** *Kids (under 10)*

You found a jar of homemade jam in the freezer, but the lid popped up a little. The jam still smells sweet and looks okay. Is this safe to eat?

Why: If a jar lid pops up after freezing, the seal might be broken, which lets air in and germs can grow. But if the jam still smells sweet, looks good, and isn't moldy, it can be eaten. If it smells bad or has mold, it's not safe and should be thrown away.

Volunteer notes: Seals broken on jars that froze may allow spoilage. If the food is still cold (below 40°F) and shows no spoilage signs (bad smell, mold), it may be safely used; otherwise, discard.

Source: PNW 296 Freezing convenience foods that you've prepared at home - "If the seal has broken and the food is still frozen or cold (refrigerator temperature, 40 °F or below), it may be safely salvaged. ... all food ... should be examined carefully for spoilage before use."

FZ-W01 **SAFE** *Tweens (10-14)*

You froze cooked stew right after it cooled down by placing the pot in an ice-water bath and then put it in the freezer. You stored it at 0°F (-18°C) all along until you used it months later.

Why: Cooling leftover foods quickly before freezing helps stop bacteria from growing. Keeping the freezer at 0°F or below keeps food safe and preserves quality for months. Food kept at this temperature is safe to eat even after long storage.

Volunteer notes: Freezing food at 0°F or below keeps bacteria from growing, though freezing doesn't kill most bacteria. Quick cooling before freezing prevents foodborne illness. Stored properly in the freezer, cooked foods remain safe for months. Follow temperature guidelines and cool large pots before freezing.

Source: PNW 214 Freezing fruits and vegetables - "Store them at 0°F (-18°C) or below. ... long-term freezer storage will affect the quality of the frozen foods, but they will be safe to eat."

FZ-W02 **NOT SAFE** *Tweens (10-14)*

You found a can of canned beans in your unheated garage that froze over the winter. The can is bulging and the lid looks like it might be broken.

Why: If a can is bulging or the lid is broken, bacteria might have grown inside even if it was frozen. Eating food from damaged cans can cause food poisoning, so it should be thrown away.

Volunteer notes: Frozen canned foods can be safe if seams are intact. Bulging lids or broken seams indicate unsafe food due to bacterial growth risk. Discard any cans with bulges, leaks, or broken seals to prevent botulism or foodborne illness.

Source: SP 50-695 Safety of canned food that freezes - "If the seam has broken, and/or the lids are bulging, and the food has thawed to room temperature, it should be discarded."

FZ-W03 IT DEPENDS *Tweens (10-14)*

You thawed frozen meat in the fridge, but it sat there for 3 days before you cooked it.

Why: Thawing meat in the fridge is safe if you cook or freeze it within a couple of days. After 3 days, the meat might start to spoil and could make you sick. Always check the meat for smell or color changes before cooking.

Volunteer notes: Meat thawed in the refrigerator should be cooked within 1-2 days to ensure safety. Longer storage increases risk of bacteria growth. If unsure about freshness, discard to avoid foodborne illness. Always keep the fridge below 40°F.

Source: PNW 296 Freezing convenience foods that you've prepared at home - "Foods that have been frozen and thawed require the same care as fresh foods. Thawed ground meat, poultry, or fish that have an off-odor or off-color should not be used."

FZ-T01 SAFE *Teens (14-18)*

You freshly cook a batch of vegetable soup and cool it quickly by placing the pot in an ice-water bath, stirring often. Then, you pack the cooled soup into freezer-safe containers, leaving some headspace, and freeze it right away at 0°F (-18°C).

Why: Quick cooling stops bacteria from growing before freezing. Freezing at 0°F keeps the soup safe for months by halting bacteria growth. Leaving headspace lets the soup expand when frozen, preventing container breakage. Always use freezer-safe containers.

Volunteer notes: OSU/USDA standards say cool hot foods quickly (e.g., ice-water bath) before freezing, keep freezer at 0°F or below, leave headspace in containers to allow for expansion. Bacteria stop growing at freezing temperatures but are not killed immediately; rapid cooling is critical to safety.

Source: PNW 296 Freezing convenience foods that you've prepared at home - "Recool large pots of hot food in a container of cold water or ice water. Stir the food and replace the cold water frequently over a 30-minute period. Freeze only high-quality foods and handle them as little as possible during processing. Store frozen foods at 0°F (-18°C) or below."

FZ-T02 NOT SAFE *Teens (14-18)*

You place raw ground meat directly into the freezer without packaging or chilling it first, then find it frozen solid days later. Before cooking, you thaw it on the kitchen counter overnight at room temperature.

Why: Freezing raw meat without packaging can allow freezer burns and contamination. Thawing meat on the counter lets harmful bacteria grow rapidly at room temperature. Meat must be thawed safely in the fridge or cold water, never at room temperature.

Volunteer notes: Freezing doesn't kill bacteria, it just stops growth. Unsafe thawing in the temperature danger zone (40°F-140°F) allows bacteria to multiply. Always thaw in the refrigerator or use cold water, changing it often, per food safety guidelines.

Source: PNW 296 Freezing convenience foods that you've prepared at home - "When foods are frozen, most bacteria just stop growing but are not killed. Once thawed, bacteria will grow if food is held above 40°F. Thawed ground meat that has an off-odor or off-color should not be used. Do not thaw meat on the counter."

FZ-T03 IT DEPENDS *Teens (14-18)*

You froze some leftover cooked rice in a shallow container but left it in the fridge overnight before freezing it. Sometimes you put frozen veggies in the fridge a bit before cooking, but other times you cook them right from the freezer.

Why: Leftover cooked rice must be cooled quickly and frozen soon to stay safe because it can grow bacteria that cause food poisoning. Thawing frozen veggies in the fridge is safe, as is cooking them straight from frozen. The safety depends on how long the rice stayed in the danger zone before freezing and vegetable acidity.

Volunteer notes: Cooked rice cools slowly and can host Bacillus cereus bacteria that survive cooking and grow if cooling is slow. Rapid cooling and prompt freezing are key for safety. Vegetables' safety depends on acidity; low-acid veggies can grow bacteria if thawed too long. Follow storage and thawing recommendations carefully.

Source: PNW 214 Freezing fruits and vegetables - "Store frozen foods at 0°F (-18°C). It is safe to refreeze fruits and vegetables that still contain ice crystals. If the temperature has warmed above 40°F, foods may not be safe. Low-acid vegetables can support bacterial growth if thawed improperly."

FZ-A01 **SAFE** *Adults*

Marcy freezes a batch of homemade vegetable soup immediately after cooking, cooling it quickly in an ice bath before portioning and placing it in airtight containers in her freezer held at 0°F (-18°C). She always labels the containers with the date and plans to use them within 3 months.

Why: Freezing cooked soup right after rapid cooling prevents harmful bacterial growth. Using airtight containers and a freezer at 0°F (-18°C) maintains safety and quality. Labeling with dates helps with proper food rotation and timely use to maintain quality.

Volunteer notes: OSU/USDA standards recommend rapid cooling before freezing, storing at 0°F or below, and labeling foods for best quality and safety. Proper handling minimizes risk of bacterial growth and spoilage.

Source: PNW 296 Freezing convenience foods that you've prepared at home - "Store them at 0°F (-18°C) or below. Use a freezer thermometer to monitor the temperature. ... Freezing convenience foods 7"

FZ-A02 **NOT SAFE** *Adults*

Tom stores canned vegetables in his unheated garage over winter where temperatures drop below 30°F. He notices some cans have bulging lids but thaws them and uses the contents without further inspection or heating.

Why: When canned foods freeze and cans bulge or seams break, the risk of spoilage increases. Using thawed canned foods with bulging or damaged cans without further heat treatment can expose you to dangerous bacteria like Clostridium botulinum. Such foods should be discarded or boiled 10 minutes before using if still safe physically.

Volunteer notes: USDA and OSU guidelines warn against using canned foods with bulging lids after freezing unless properly boiled. Freezing can damage cans leading to contamination. Always inspect canned goods after freezing and cook low-acid foods thoroughly.

Source: PNW 296 Freezing convenience foods that you've prepared at home - "If the seam has broken, and/or the lids are bulging, and the food has thawed to room temperature, it should be discarded. For extra safety, boil low acid foods for 10 minutes before using."

FZ-A03 **IT DEPENDS** *Adults*

Lena thawed frozen green beans that had partially warmed to 45°F during a power outage. She wonders if she can safely refreeze or cook them.

Why: Safety depends on how long the beans were above 40°F. If they stayed above 40°F for less than 2 hours and still contain ice crystals, refreezing is generally safe though quality may suffer. Longer warm exposure can allow bacteria to grow, making refreezing unsafe. Cook immediately if unsure, or discard if suspect.

Volunteer notes: Frozen vegetables are safe if still mostly frozen or kept below 40°F. At temperatures above 40°F for extended times, bacteria can grow. Use a freezer thermometer and assess time above 40°F to decide. Safety depends on time, temperature, and condition of food.

Source: PNW 214 Freezing fruits and vegetables - "If the temperature has warmed above 40°F (5°C), foods may not be safe for refreezing. Thawed vegetables above 40°F can grow harmful bacteria and may not show spoilage."

DEHYDRATING & SMOKING

DS-K01 **SAFE** *Kids (under 10)*

You helped your family dry apples by cutting them into thin slices and watching them in the dehydrator until they were dry and leathery. Now the apple slices are stored in a clean, airtight container on the kitchen shelf.

Why: Drying apples until they are leathery means most water is gone, so bad germs can't grow. Keeping them in a clean, airtight place stops new germs from getting on them. This way, your dried apples stay safe to eat!

Volunteer notes: Proper dehydration removes moisture that bacteria need to grow. Store dried fruits in airtight containers at room temperature to prevent moisture reabsorption and spoilage. This follows PNW 397's guidance on drying fruits properly.

Source: PNW 397 Drying fruits and vegetables - "Drying is a method of food preservation in which moisture is removed from fruits with a dehydrator or other drying methods to make them shelf stable."

DS-K02 **NOT SAFE** *Kids (under 10)*

You smoked some fish outside, but when you opened the smokehouse later, the fish smelled strong and funny, and the outside was sticky and slimy.

Why: A bad smell and sticky, slimy fish means germs might be growing. If fish smells funny or looks slimy, it can make you sick. It's best to throw it away instead of eating it.

Volunteer notes: Smoked fish must be properly smoked and cooled; off-odors and slimy texture indicate spoilage or unsafe microbial growth, making it unsafe to consume (PNW 238 Smoking fish at home).
Source: PNW 238 Smoking fish at home - "Signs of spoilage like off-odor and sticky, slimy texture indicate unsafe fish after smoking."

DS-K03 **IT DEPENDS** *Kids (under 10)*

You left beef jerky out on the kitchen counter all day after drying it. It looks dry but smells a little different than when you first made it.

Why: Jerky is safe when it is dried well and stored in a cool, dry place. But if it smells bad or looks moldy, it can make you sick. Watch for bad smells or fuzzy spots-that's when it's not safe.
Volunteer notes: Jerky safety depends on drying to the correct moisture level and storage conditions. If jerky is left out too long or smells off, it may have unsafe bacteria or mold growth (PNW 632 Making jerky at home safely).
Source: PNW 632 Making jerky at home safely - "Proper drying and storage are key; watch for off smells or mold for safety."

DS-W01 **SAFE** *Tweens (10-14)*

Maria dried apple slices using a home dehydrator. She made sure the apples were fully dried and stored them in an airtight container in a cool, dark place.

Why: If fruit is properly dried so it has very little moisture left, and stored in a cool, dark place that prevents moisture and light, it stays safe to eat. This stops mold and bacteria from growing.
Volunteer notes: Fully dried fruit stored in proper conditions prevents spoilage and microbial growth, ensuring safety. Moisture control and storage temperature are critical. Cool, dark, low-moisture storage is recommended by PNW 397.
Source: PNW 397 Drying fruits and vegetables - "Pretreating helps to stop enzyme action; storing dried fruit at cool temperatures helps prevent browning and spoilage; fully dried fruit stored properly is safe."

DS-W02 **NOT SAFE** *Tweens (10-14)*

James smoked fish in a small metal smoker for 2 hours but did not finish cooking the fish in the oven afterward.

Why: Small metal smokers often can't reach high enough temperatures to kill harmful bacteria. If fish isn't cooked to 160°F after smoking, it might still have dangerous germs that can make you sick.
Volunteer notes: To ensure smoked fish safety, it must be cooked to a safe internal temperature (160°F). Metal smokers alone may not be sufficient; oven cooking to finish is required to kill pathogens, as per PNW 238 standards.
Source: PNW 238 Smoking fish at home - "Small metal smokers should be supplemented by oven cooking to achieve a core temperature of 160°F to ensure safety."

DS-W03 **IT DEPENDS** *Tweens (10-14)*

Alex made beef jerky using a home dehydrator but did not heat the meat before drying. He wondered if the jerky was safe to eat.

Why: Jerky made safely requires the meat to be heated to 160°F before or right after drying to kill germs. If the meat wasn't heated properly, it might still contain harmful bacteria. So, safety depends on whether the meat reached that temperature.
Volunteer notes: Jerky safety depends on heating meat to 160°F to destroy pathogens like Salmonella or E. coli. Drying alone isn't enough. Conditioning steps or precooking are needed for safe jerky per PNW 632 guidance.
Source: PNW 632 Making jerky at home safely - "Jerky is safe only when meat has been heated sufficiently to 160°F before or immediately after dehydration to destroy pathogens."

DS-T01 **SAFE** *Teens (14-18)*

I use an electric dehydrator with a built-in thermostat set to 155°F to dry beef strips for jerky. After drying, I immediately check the temperature and confirm it reached at least 160°F for safety before storing.

Why: Jerky is only safe if the meat reaches 160°F to destroy harmful bacteria such as Salmonella and E. coli. Using a dehydrator that keeps a steady temperature of 145°F-155°F and verifying the final internal temperature ensures safety. This method prevents foodborne illness and allows jerky to be shelf-stable for weeks.
Volunteer notes: USDA and university research requires heating meat to 160°F (165°F for poultry) before or just after drying to destroy pathogens. Dehydrators must maintain correct temperature to be safe. Checking temperature with a thermometer is essential to confirm doneness.
Source: PNW 632 Making jerky at home safely - "Jerky can be considered "done" and safe to eat only when meat has been heated sufficiently to 160°F and poultry to 165°F before or immediately after the dehydration process. This destroys any pathogens present and creates jerky that is dry enough to be shelf-stable."

DS-T02 **NOT SAFE** *Teens (14-18)*

I smoke fish at home by hanging it outside in an unheated, smoky area without controlling the temperature or time, just guessing when it looks 'done.'

Why: Smoking fish without controlling temperature or time can allow dangerous bacteria or parasites to survive. Unlike proper cooking, cold or uncontrolled smoking does not reliably kill pathogens, risking foodborne illness such as from Clostridium botulinum or parasites like Trichinella.

Volunteer notes: Safe smoking requires specific temperatures and times to kill pathogens and parasites. Uncontrolled outdoor smoking may keep food in the 'danger zone' temperature where bacteria grow rapidly. Smoking alone doesn't always make fish safe.

Source: PNW 238 Smoking fish at home - "The Trichinella parasite is dangerous too... For illness to occur, the pathogen must survive the process and be consumed. Smoking processes must reach temperatures to kill these pathogens safely."

DS-T03 **IT DEPENDS** *Teens (14-18)*

I'm drying slices of garden tomatoes which are acidic, but I'm not sure how long to dry them or if I need to heat them before storage.

Why: Drying acidic fruits like tomatoes is usually safer because harmful bacteria grow less in acid. However, safety depends on how long you dry them, the drying temperature, and finishing steps. Tomatoes need to be dried until moisture is low enough to prevent spoilage, and if not dried properly, can still grow mold or bacteria.

Volunteer notes: Low-acid foods (like meat) require higher heat or cooking; high-acid foods (like tomatoes) are less risky but must be dried thoroughly. Time, temperature, and final moisture level determine safety, so incomplete drying or improper handling can cause spoilage.

Source: PNW 397 Drying fruits and vegetables - "Acidic foods include all fruit, tomatoes, and pickles made with an adequate amount of vinegar... Low acid foods require more care to prevent pathogens. Proper drying until moisture is low enough is critical."

DS-A01 **SAFE** *Adults*

You use a food dehydrator that maintains a steady temperature of 145°F to dry beef jerky. You always check the internal temperature of the dehydrator with an accurate thermometer before starting, ensuring it reaches the required temperature.

Why: Drying jerky safely requires drying meat at a temperature of at least 145°F to kill pathogens. Using a dehydrator with a thermostatically controlled temperature dial and confirming with an external thermometer helps ensure the safety of the dried product.

Volunteer notes: Jerky must be dried at 145°F or above to be safe; dehydrators must have thermostats to maintain and verify this temperature. This practice prevents growth of harmful bacteria in low-moisture meat products.

Source: PNW 632 Making jerky at home safely - "Use a thermometer to determine that the empty, operating dryer can maintain a temperature of at least 145°F. Do not use dehydrators with factory preset temperatures that cannot be controlled."

DS-A02 **NOT SAFE** *Adults*

You smoke fish at home but do not monitor the temperature or time during the smoking process. You finish when the fish looks dried without measuring internal temperature or following any tested recipe.

Why: Smoking fish without monitoring internal temperature and time can allow harmful bacteria to survive, particularly Clostridium botulinum. Uncontrolled smoking is risky and can result in foodborne illness.

Volunteer notes: Smoking fish must follow tested recipes and include proper temperature and time controls to prevent botulism. Without these controls, fish can remain unsafe due to toxin risk.

Source: PNW 238 Smoking fish at home - "Smoking fish requires careful control of time and temperature to prevent botulism toxin production."

DS-A03 **IT DEPENDS** *Adults*

You are drying tomatoes using a homemade solar dryer on your covered porch. You are unsure if the air temperature consistently reaches the recommended drying temperature and you don't know the altitude of your location.

Why: The safety of drying tomatoes depends on ensuring the drying temperature is high enough (usually above 130°F) for a sufficient time to reduce moisture safely. Altitude affects drying times and temperatures needed. Without knowing these details or controlling temperature, safety cannot be assured.

Volunteer notes: Drying depends on temperature, drying time, and altitude. Low temperatures or high altitude may require longer drying to prevent spoilage. Check air temperature and adjust time accordingly for safety.

Source: PNW 397 Drying fruits and vegetables - "Drying times vary by altitude and temperature; foods must reach safe moisture levels to be shelf stable."